AI in College

A Tool for Learning, Not a Shortcut Around Thinking

David Geddam

Bob Jones University

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Why AI

- Why I care about AI as a computer science student.
- A quick example from my own study routine.

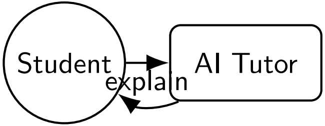
The Ethical Problem

Student 1

"Explain this concept step by step so I can understand it."

Student 2

"Write the whole assignment so I can submit it."



Fun Fact

Same tool. Same classroom. Same deadline. Completely different ethical choice.

Proposition and Standard of Judgment

Proposition

College students should be allowed to use AI tools for academic work, within clear ethical boundaries.

Responsible academic AI use

Using AI to support learning, understanding, organization, brainstorming, and revision while preserving the student's own effort, honesty, authorship, and critical thinking.

Standard 1 Strengthen learning, not replace learning.

Standard 2 Preserve academic integrity through honesty and disclosure.

Source: UNESCO, Guidance for Generative AI in Education and Research; EDUCAUSE, A Framework for Transparent GenAI Use in Higher Education.

What Is an AI Model? From Inputs to Outputs

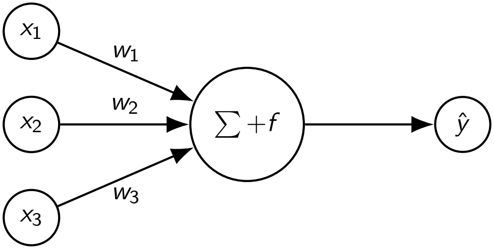
Input text → Tokens → Embeddings → Neural layers → Output

- A large language model does not simply "look up" answers like a search engine.
- It learns patterns in language from data, then predicts likely next tokens.
- This power can support learning, but it can also produce fluent mistakes.

Fun Fact

Prediction is powerful, but prediction is not the same thing as perfect understanding.

A Neuron: The Smallest Building Block



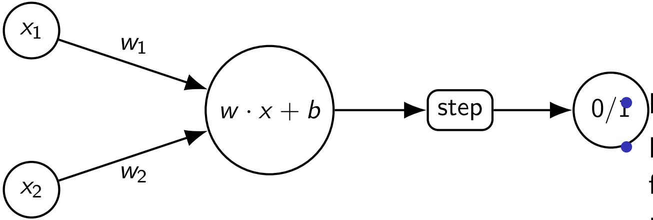
Math idea:

$$z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

$$\hat{y} = f(z)$$

- x_i: inputs
- w_i: learned weights
- b: bias
- f: activation function

Perceptron: Learning a Decision Boundary



$$\hat{y} = \begin{cases} 1, & w \cdot x + b \geq 0 \\ 0, & w \cdot x + b < 0 \end{cases}$$

- Early neural network idea.
- Learns weights that separate one class from another.
- Modern deep networks stack many layers and learn much richer patterns.

How Learning Happens: Loss and Gradient Descent

Training goal

Make the model's prediction $\hat{y}$ close to the correct answer y.

$$L(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell(f_\theta(x_i), y_i)$$

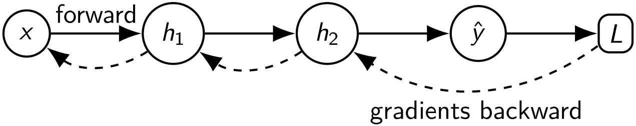
$$\theta_{new} = \theta_{old} - \eta \nabla_\theta L(\theta)$$

- θ: model parameters / weights
- L: loss, or how wrong the model is
- η: learning rate
- $\nabla_\theta L$: direction that tells weights how to change

Fun Fact

Training is basically: predict, measure error, adjust weights, repeat.

Backpropagation: How Error Moves Backward



$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h_2} \frac{\partial h_2}{\partial h_1} \frac{\partial h_1}{\partial w}$$

- Backpropagation uses the chain rule from calculus.
- It tells each weight how much it contributed to the error.

Transformer

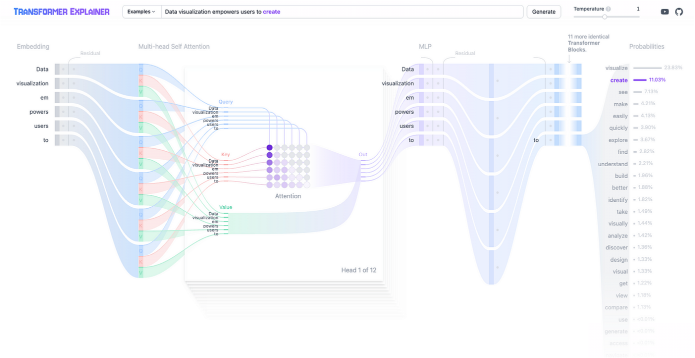


Figure: High-level Transformer architecture showing how self-attention helps the model understand relationships between words.

Source: Vaswani et al., Attention Is All You Need, 2017.

Self-Attention: The Key Math Idea

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Q = Query: what am I looking for?
- K = Key: what information is available?
- V = Value: what information should I use?
- $\sqrt{d_k}$ keeps scores numerically stable.

Example

"The student used AI because he was confused."

- The word "he" should attend to "student."
- The model learns word relationships from context.

Fun Fact

Self-attention lets the model decide which words should pay attention to which other words.

Black-Box Problem: We Know the Math, But Not Every Decision

What we understand:

- tokens and embeddings
- attention formula
- loss functions
- backpropagation
- gradient descent

What is still hard:

- why one exact answer was chosen
- what internal features mean
- when confidence is misleading
- whether the model truly understands

Important

Because AI is powerful but partly opaque, colleges should teach verification, disclosure, and critical thinking — not blind trust.

Dangers of AI in Academic Work

Learning dangers

- hallucinated facts
- fake citations
- wrong code or math
- shallow understanding
- overdependence

Ethical dangers

- hidden AI use
- submitting AI-written work
- weakening authorship
- unfair grading
- academic dishonesty

Fun Fact

The danger is not only that AI can be wrong. The danger is that AI can be wrong with confidence.

Source: APA, Teaching academic writing in the age of AI, 2026.

Responsible Use: Tutor, Not Substitute

Acceptable

- brainstorm ideas
- explain hard concepts
- summarize readings
- create practice questions
- revise grammar
- debug understanding

Unacceptable

- submit AI-written work as your own
- fake citations
- use AI on closed-book exams
- hide major AI assistance
- replace personal effort

Example disclosure

"I used AI to brainstorm ideas and revise grammar, but the final argument and writing are my own."

World Models: Could AI Understand a Student's Learning Process?

Definition

A world model is an AI system's internal representation of how things work, how actions lead to consequences, and how situations change over time.

Attempts → Mistakes → Revisions → Understanding

- Future AI may help instructors analyze process, drafts, mistakes, and improvement.
- But AI should help review learning, not become automatic punishment.

AI Cheating Detection: Helpful, But Not Judge and Jury

AI might flag:

- sudden style changes
- no draft history
- polished work with no understanding
- suspicious patterns across submissions

But fairness requires:

- human review
- student explanation
- transparent rules
- no automatic punishment

AI safety connection

The more powerful AI becomes, the more important alignment with human values becomes: fairness, honesty, accountability, and learning.

Conclusion

Review

College students should be allowed to use AI tools for academic work, within clear ethical boundaries.

1. AI can support learning when used as a tutor, not a substitute.
2. AI has limitations: hallucinations and black-box behavior.
3. Clear boundaries are better than fear or total bans.
4. Future AI may help education, but it must be fair and human-reviewed.

Fun Fact

AI should not replace the student's mind. It should challenge the student's mind to become stronger.

Works Cited / Verbal Sources

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Thank you!